

Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2025

Course: B.Tech
Code: 105301

Subject: Digital Electronics

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. **1** is compulsory.

Q.1 Choose the correct option / answer the following (Any seven questions only)

[2 x 7 = 14]

- (a) A Programmable Logic Array consists of
 - (i) Fixed AND and fixed OR arrays
 - (ii) Programmable AND and programmable OR arrays
 - (iii) Fixed AND and programmable OR arrays
 - (iv) Programmable AND and fixed OR arrays
- (b) The invalid condition in an SR flip-flop occurs when
 - (i) $S = 0, R = 0$
 - (ii) $S = 0, R = 1$
 - (iii) $S = 1, R = 0$
 - (iv) $S = 1, R = 1$
- (c) A ripple counter is also known as
 - (i) Synchronous counter
 - (ii) Up-down counter
 - (iii) Asynchronous counter
 - (iv) Ring counter
- (d) The resolution of an n-bit DAC is
 - (i) 2^n
 - (ii) $1/(2^n - 1)$
 - (iii) $1/2^n$
 - (iv) 2^{n-1}
- (e) Which memory is volatile in nature?
 - (i) ROM
 - (ii) RAM
 - (iii) PROM
 - (iv) EPROM
- (f) Find the 2's complement of the binary number 1010
 - (i) 0101
 - (ii) 0110
 - (iii) 1110
 - (iv) 0111
- (g) Which logic family is most suitable for battery-operated portable devices, and why?
 - (i) TTL – high speed
 - (ii) RTL – low complexity
 - (iii) CMOS – low power consumption
 - (iv) DTL – low cost
- (h) A Multiplexer (MUX) is a device that
 - (i) Converts parallel data into serial
 - (ii) Performs arithmetic operations
 - (iii) Selects one input from many and routes it to the output
 - (iv) Stores data
- (i) Evaluate the Boolean expression $Y = A + A \cdot B$
 - (i) A
 - (ii) $A + B$
 - (iii) B
 - (iv) 0
- (j) What is the output of a half adder for inputs $A = 1$ and $B = 1$?
 - (i) Sum = 0, Carry = 1
 - (ii) Sum = 1, Carry = 0
 - (iii) Sum = 1, Carry = 1
 - (iv) Sum = 0, Carry = 0

- Q.2** (a) Implement the logic expression using a suitable Multiplexer $F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$. Using variables A, C, and D as a select line. [7]
 (b) Describe the architecture and working of Programmable Logic Array. [7]
- Q.3** (a) Design a synchronous counter using a T-flip-flop that counts $0 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 0$. [7]
 (b) Describe the dual slope ADC and explain why it is suitable for digital voltmeters. [7]
- Q.4** (a) Describe the CMOS logic family and its advantages over TTL. Discuss its role in modern VLSI design. [7]
 (b) Simplify the Boolean function: $F(A, B, C, D) = \sum(0, 2, 4, 5, 6)$. [7]
- Q.5** (a) Implement a full-adder circuit with a decoder and two OR gates. [7]
 (b) Compare PLA, PAL, and ROM-based implementations. [7]
- Q.6** (a) Design the logic expression $F = A(B + CD) + BC'$, using NAND gates only. [7]
 (b) Describe charge-coupled device memory and its working principle. [7]
- Q.7** (a) Explain the working of encoders and decoders. Design a 3-to-8 decoder logic circuit. [7]
 (b) Explain the concept of memory organization and operation in digital systems. [7]
- Q.8** (a) Explain the concept of the race-around condition in the JK flip-flop and suggest methods to eliminate it. [7]
 (b) Explain the R-2R ladder DAC and compare it with the weighted resistor DAC. [7]
- Q.9** Write short notes on **any two** of the following: [7 x 2 = 14]
 (a) Flip flops
 (b) sample-and-hold circuit
 (c) PROM, EPROM and EEPROM
 (d) BCD to Gray code conversion